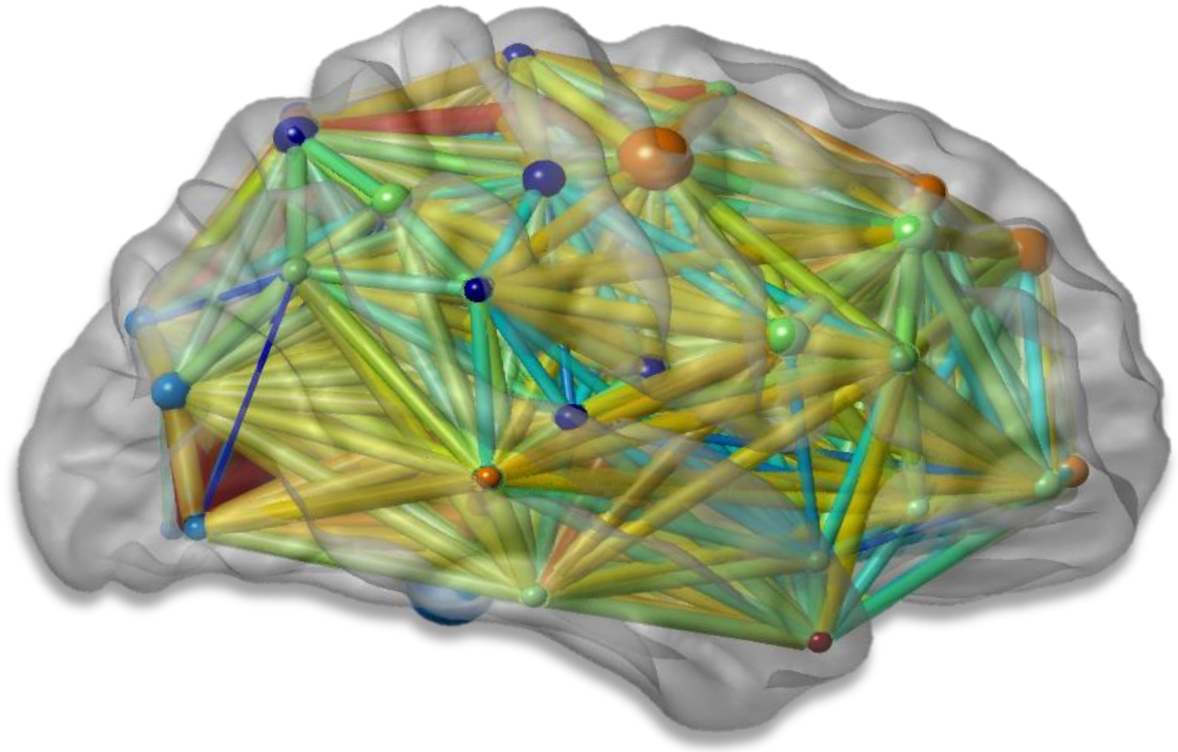


Brain Connectivity in Bipolar Disorder

Prof Dara M Cannon
University of Galway

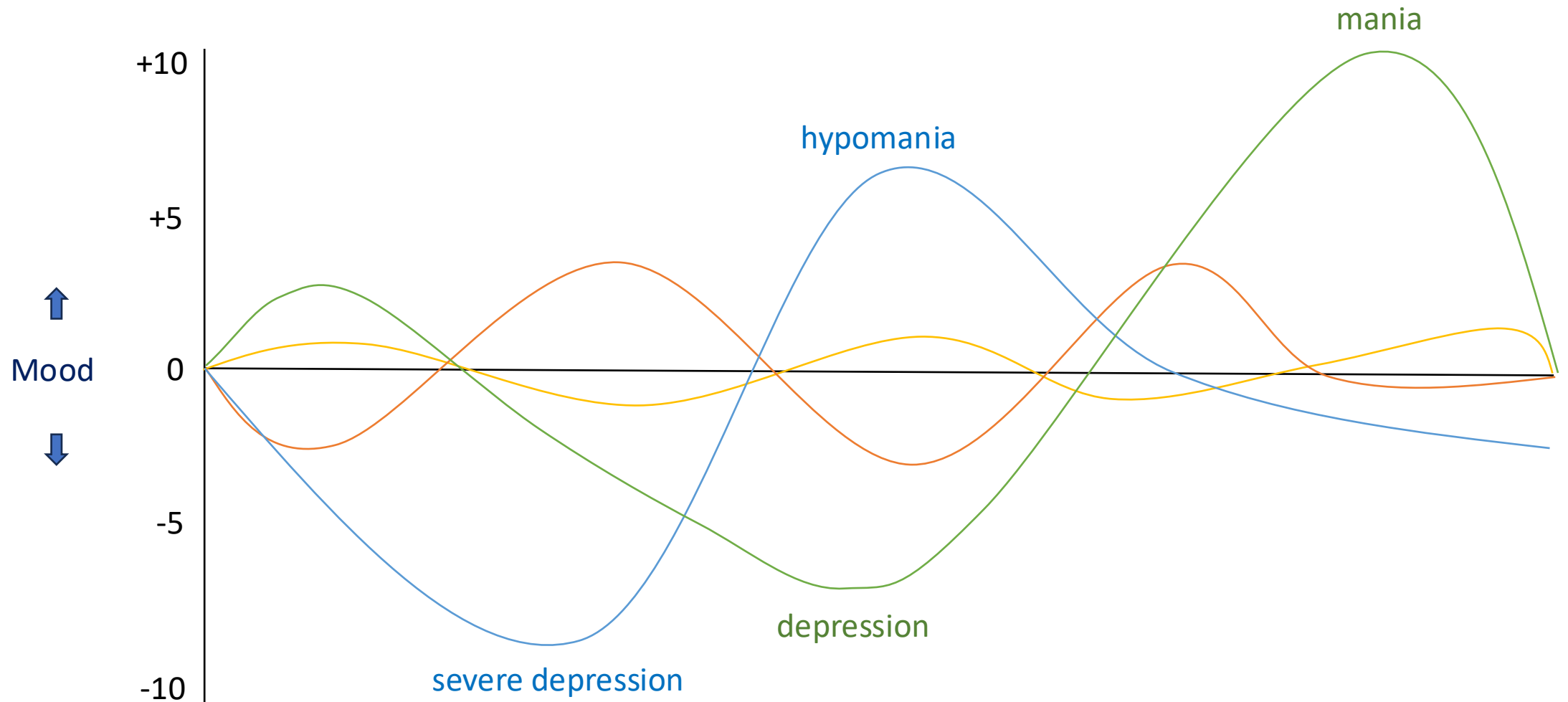
ISRC-CN3
University of Ulster
Derry

August 2026



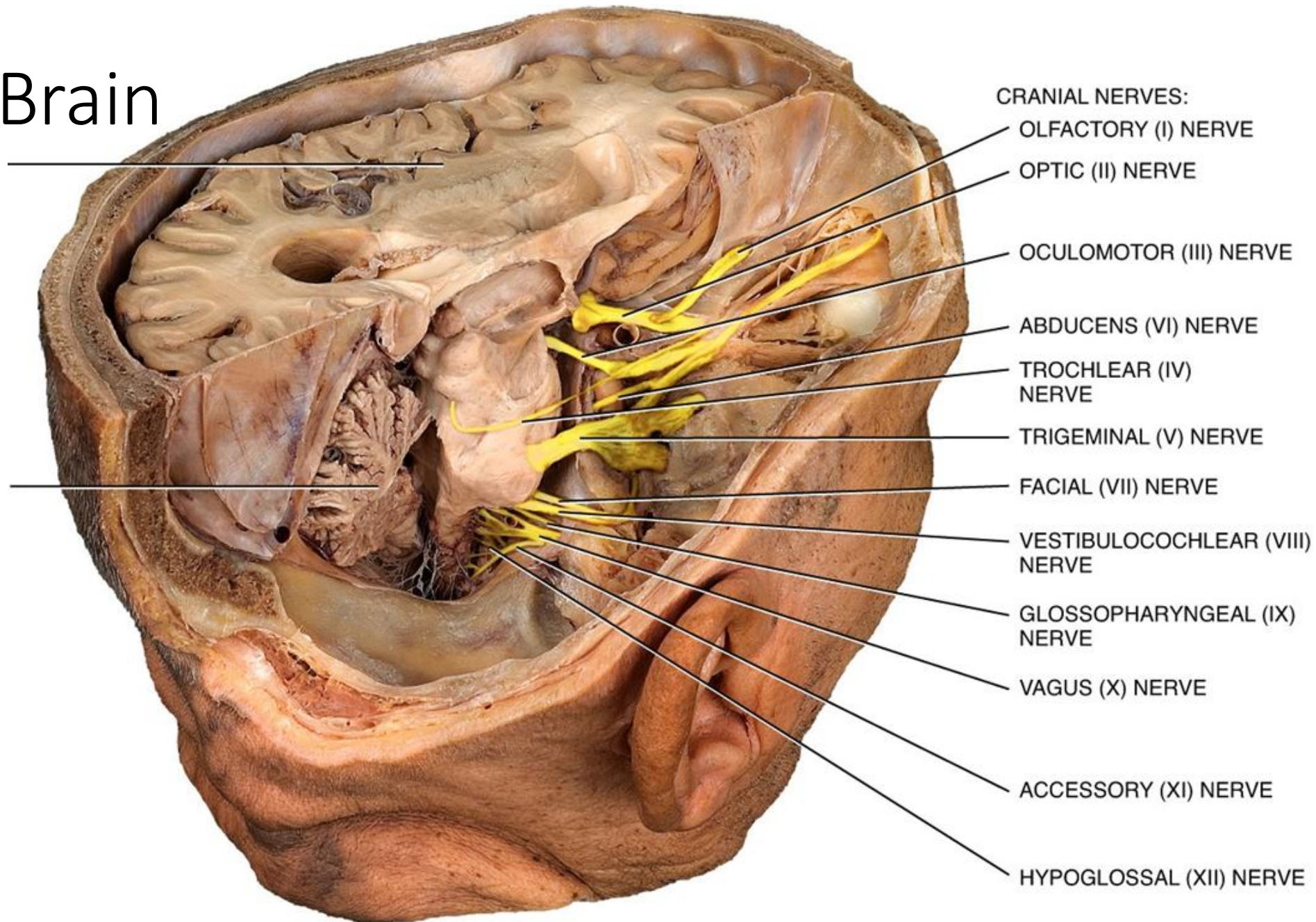
UNIVERSITY OF GALWAY

Bipolar Disorder – Type I & II



The Human Brain

- Cerebrum
- Thinking
- Introspection
- Networks
- Cranial Nerves
- Social Cognition
- Being Human
- Separateness

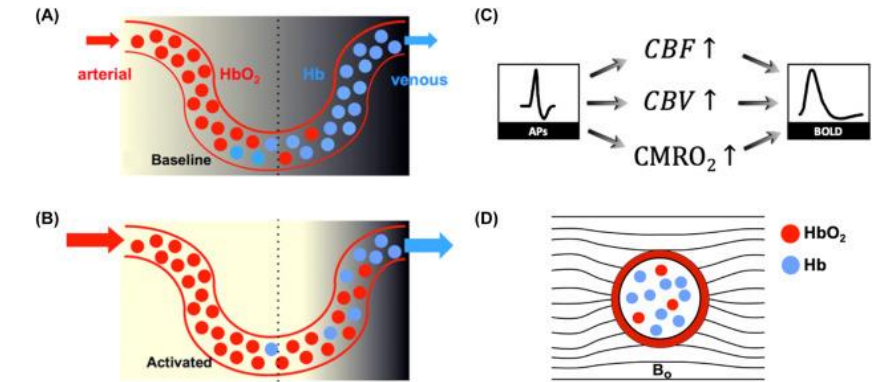
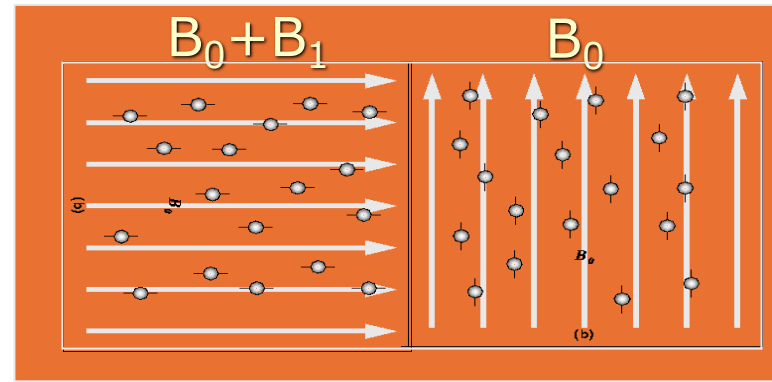
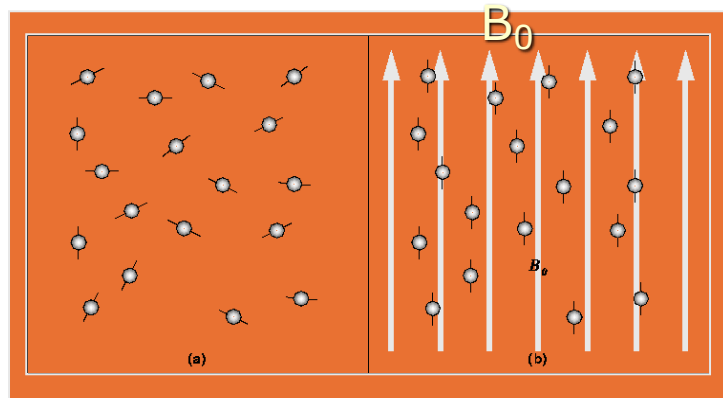


(b) Posterolateral view of dissected brain

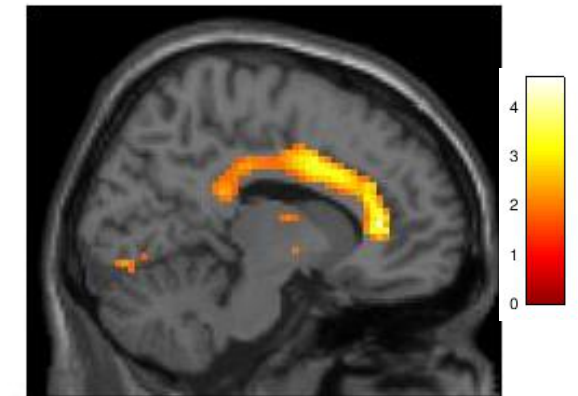
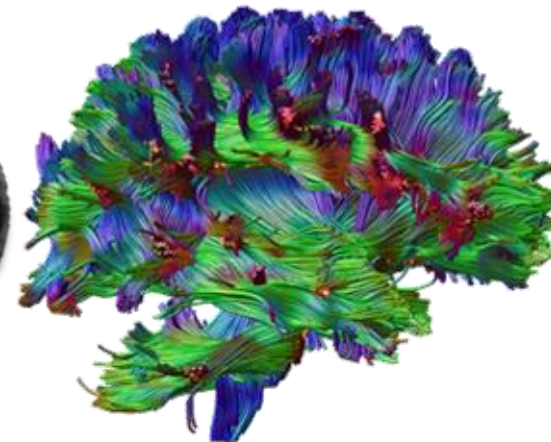
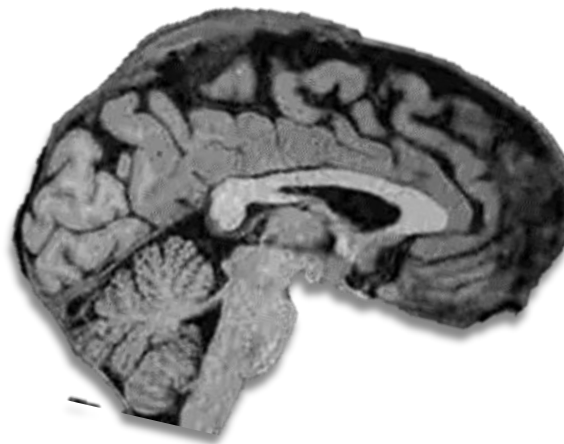
Neuroimaging – Magnetic Resonance Imaging

Structural, Diffusion & Functional MRI

Protons

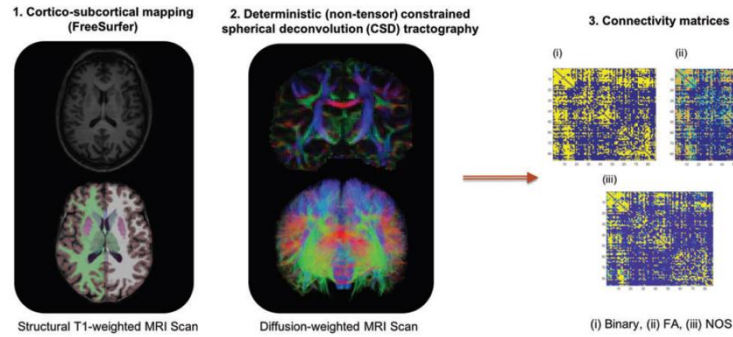


Baxter, 2023 <https://doi.org/10.1016/B978-0-323-85280-7.00002-6>



Gross Specimen | T1-weighted MRI | diffusion-weighted MRI tractography | Activation fMRI

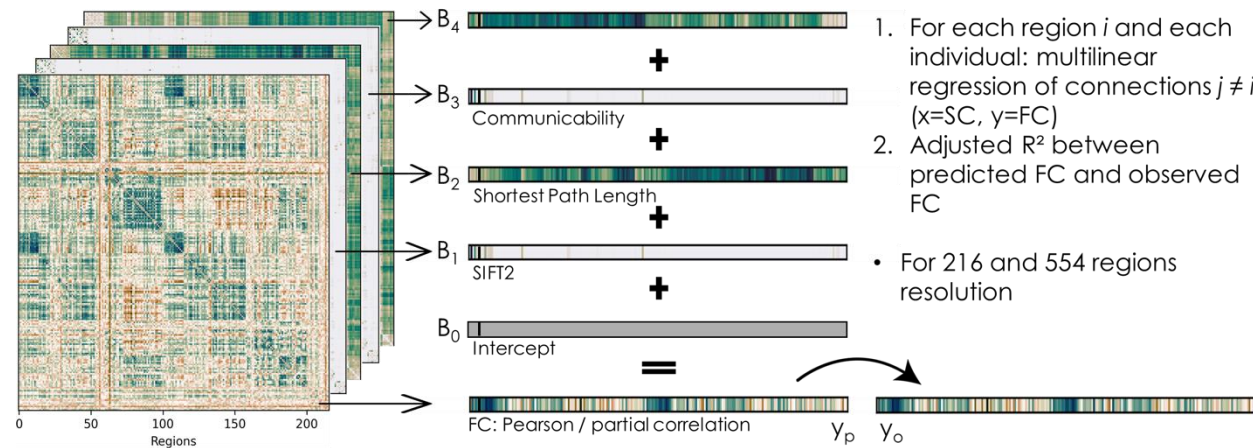
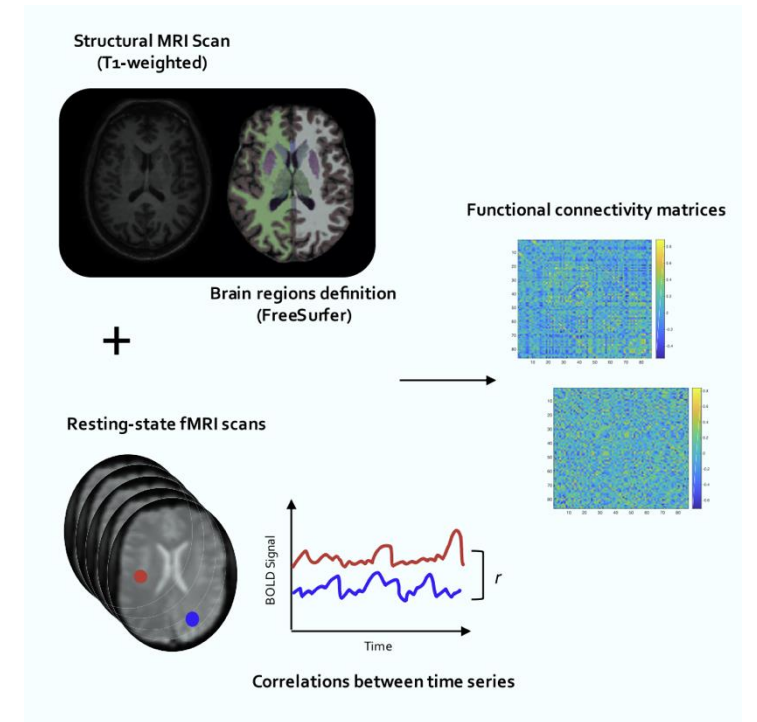
Human Brain Connectome



Structural Connectivity

Functional Connectivity

SC-FC Coupling

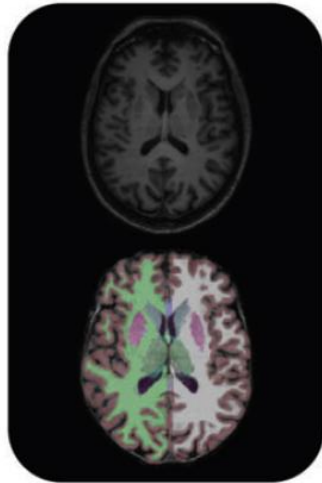


Structural Connectivity

- Topology & Subnetwork Search

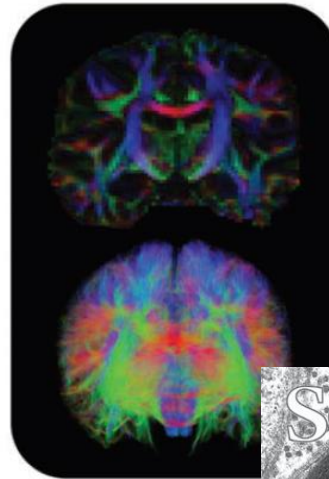


1. Cortico-subcortical mapping (FreeSurfer)



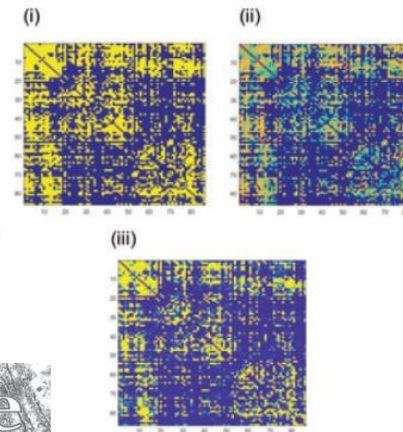
Structural T1-weighted MRI Scan

2. Deterministic (non-tensor) constrained spherical deconvolution (CSD) tractography



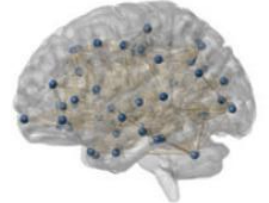
Diffusion-weighted MRI

3. Connectivity matrices



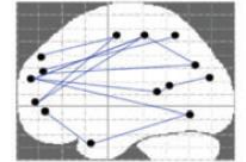
(i) Binary, (ii) FA, (iii) NOS

4. Global Network Measures



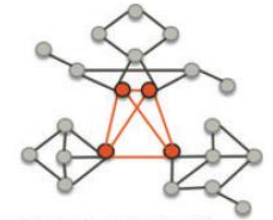
• Nodes = Grey matter regions
— Edges = White matter pathways

5. Network-based statistics



— = sub-network component

6. Rich-Club

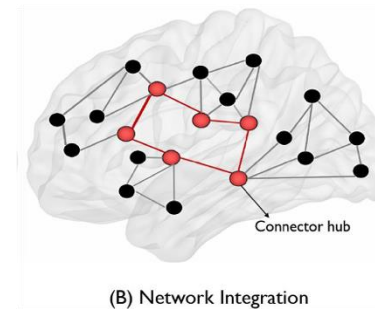
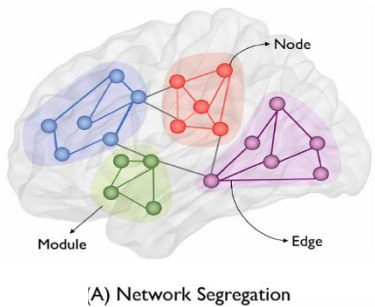


• = rich-club brain region (node)
• = non rich-club node
— = non rich-club white matter connection (edge)
— = rich-club white matter edge

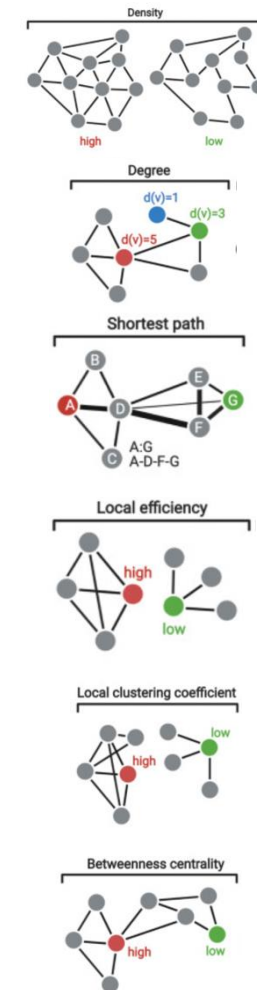
FIG. 1. Human connectome reconstruction. **(1)** Brain region definition into 34 cortical and 18 subcortical—FreeSurfer v5.3.0; **(2)** white matter trajectories—ExploreDTI v4.8.6; **(3)** connectivity matrices—ExploreDTI v4.8.6; **(4)** global network measures—Brain Connectivity Toolbox v1.52; **(5)** NBS, NBS v1.2; **(6)** rich-club analysis—Brain Connectivity Toolbox v1.52, Rstudio v1.0.14. DTI, diffusion tensor imaging; NBS, network-based statistics. Color images are available online.

Topology – Graph Measures

The **Human Connectome** is characterized by short paths, high efficiency, high clustering and high centrality

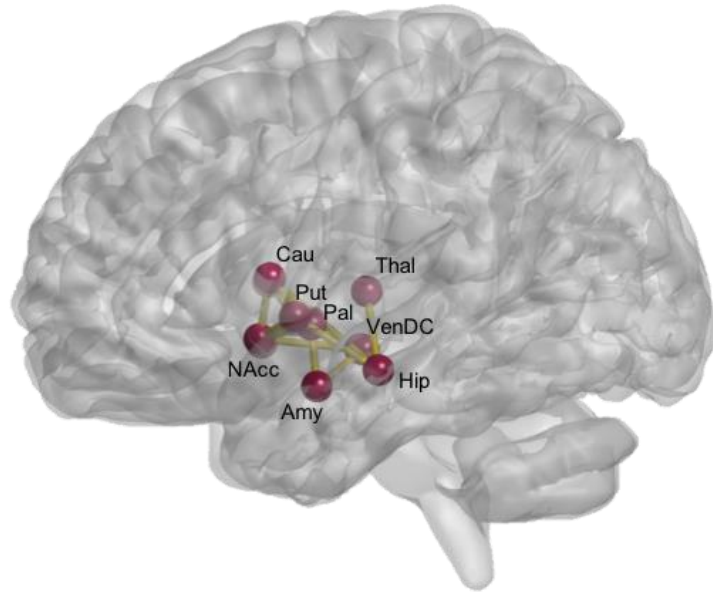


DENSITY	Estimate of overall connectivity of the network. Computed as the total number of connections present in the network over the total number of possible connections.
DEGREE	The number of all connections passing through a given node in a network.
PATH LENGTH	Estimate of most efficient paths in the network. Computed as the number of edges to be traversed to connect any pair of nodes. Commonly averaged over all nodes in the network.
EFFICIENCY	Estimate of overall communication efficiency throughout the network. Computed as average inverse shortest path between each possible pair of nodes in the graph. Commonly averaged over all nodes in the network
CLUSTERING COEFFICIENT	Estimate of local information segregation within communities, or clusters, of nodes that are highly interconnected. Computed as the average number of connections between a node and its neighbors over the total possible number of connections in a network.
CENTRALITY	A measure of how central a node or edge is to the overall network, for example in terms of the number of shortest paths that travel through the node or edge (i.e., betweenness centrality).



Structural Connectivity

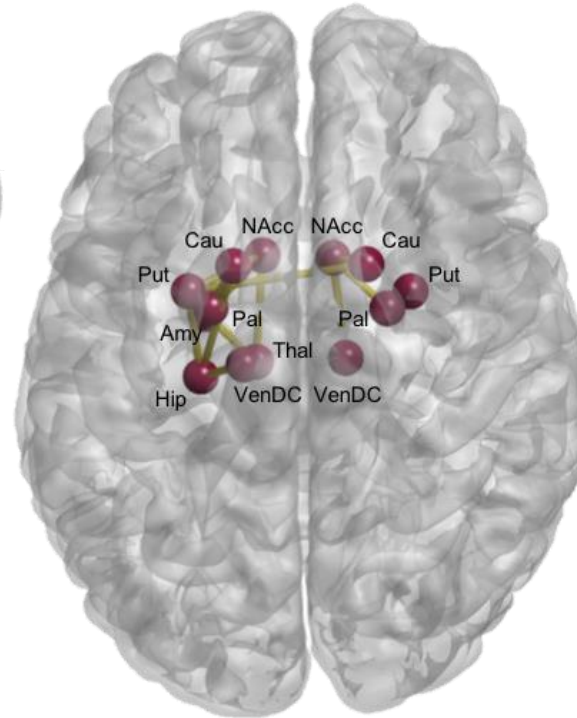
- Topology & Subnetwork Search
- Frontolimbic & basal ganglia dysconnectivity



$T > 1.5$,
 $p = 0.031$

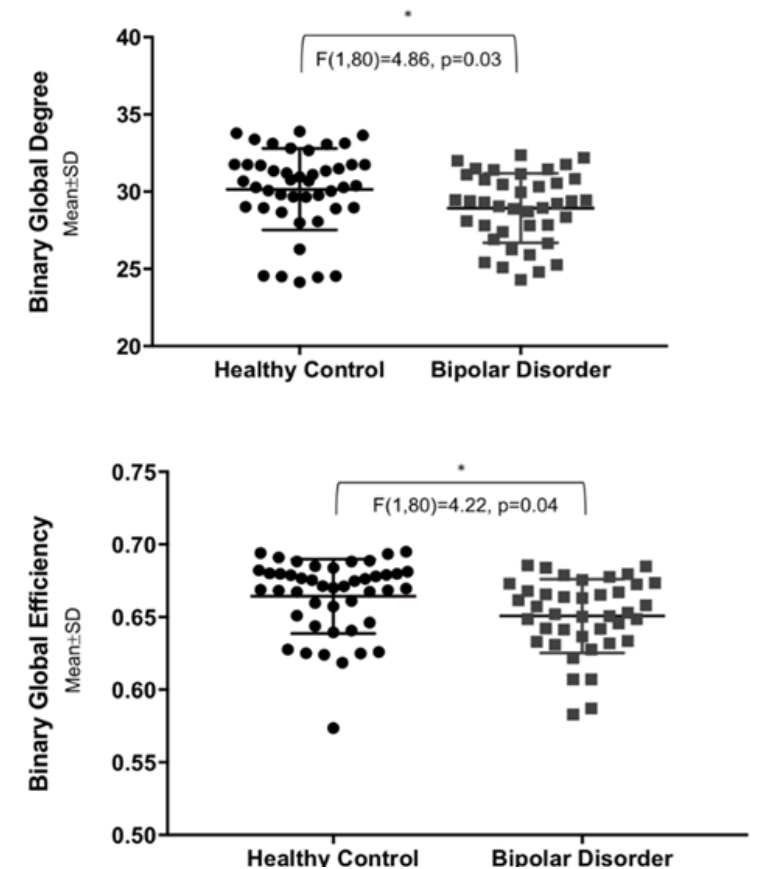
● Nodes = Grey matter regions
— Edges = White matter trajectories

Main Effect of Diagnosis
BD ($n = 33$) < HC ($n = 50$)



Main Effect of Diagnosis, BD < HC

- ↓ density of connections
- ↓ number of connections (degree)
- ↓ communication efficiency (global efficiency)



Structural Connectivity

- ENIGMA BD – 509 HC & 450 BD, 16 International Sites, 86% Euthymic
- Global Topology is Disrupted in Bipolar Disorder

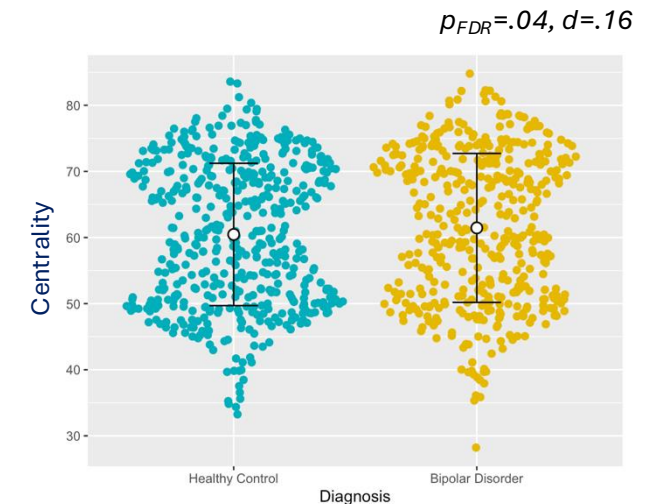
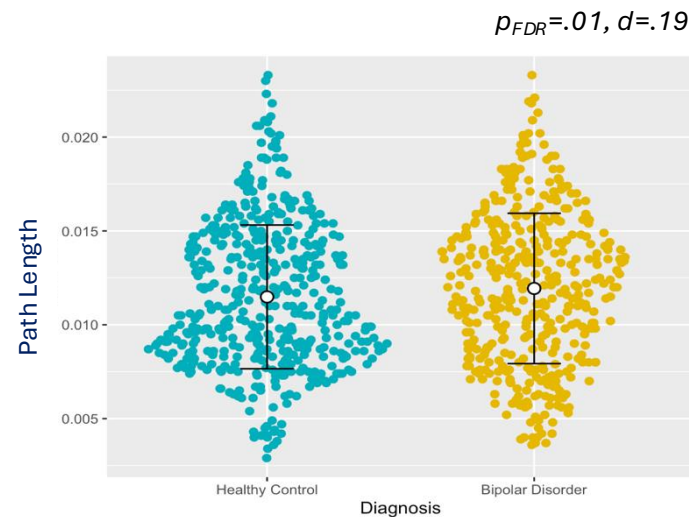
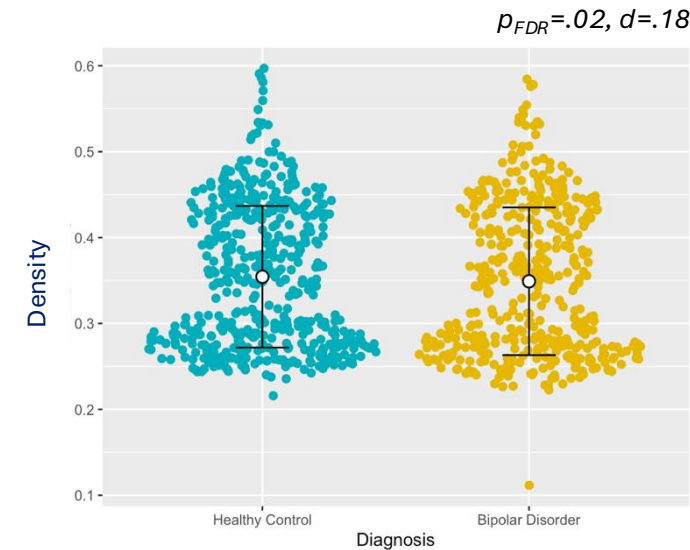
BD is associated with widespread structural connectivity changes in its brain networks



Lower connectome density, efficiency, longer paths, and higher centrality
($d=0.16-0.21$)

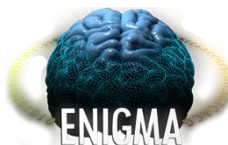


These are measures of global *integration*, that describe how easily and efficiently information can travel between different regions at the whole-brain level.



$$\text{connection}_i = \text{Intercept} + \beta_1 \cdot \text{Diagnosis} + \beta_2 \cdot \text{Age} + \beta_3 \cdot \text{Sex} + \text{random effect}(\text{site})$$

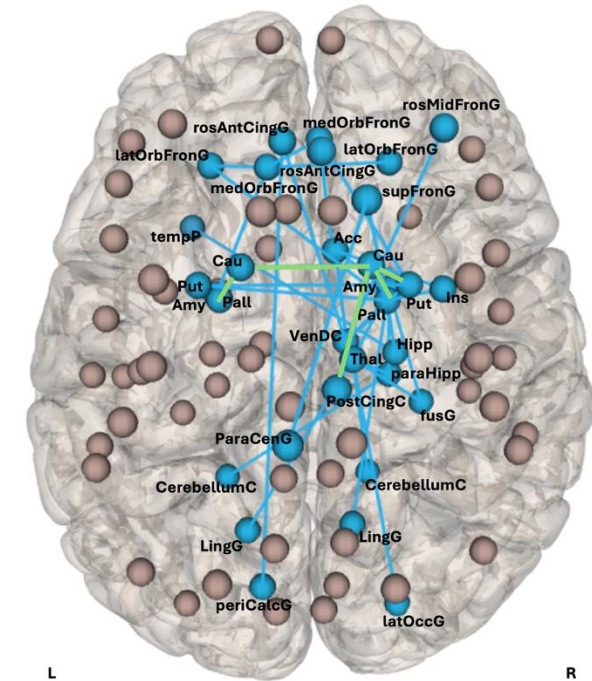
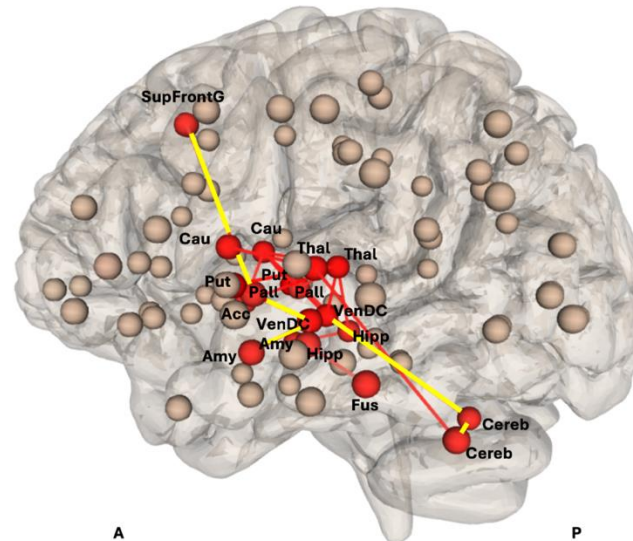
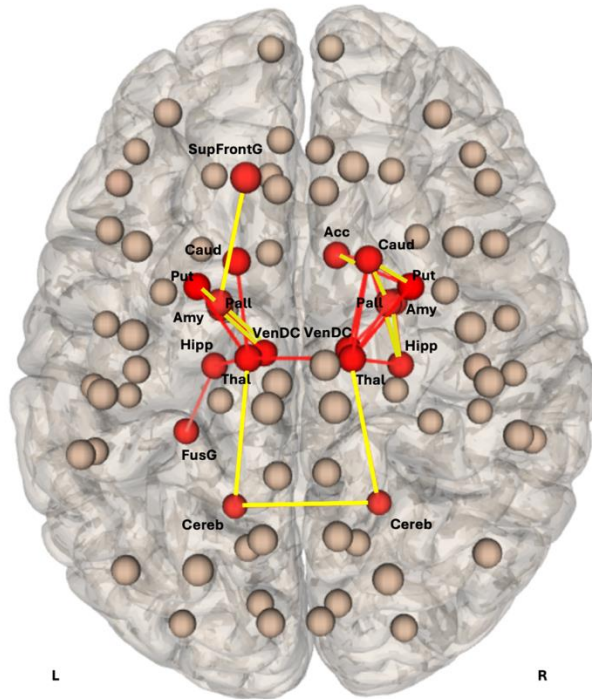
Nabulsi et al., 2026 PMID 42092437



Structural Connectivity



- Subnetwork search (NBS – Network Brain Statistic)
- ENIGMA BD – 509 HC & 450 BD, 16 International Sites, 86% Euthymic



Lower FA —
Higher FA —

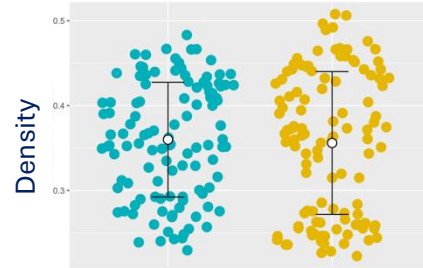
Lower streamlines count —
Higher streamlines count —

Associations with Clinical Illness Features



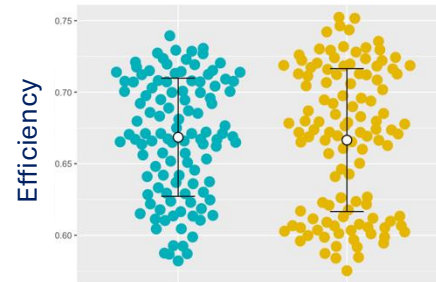
- Larger sample permits investigation of associations with a deeper phenotype than diagnosis: onset, duration ill, episodes, course severity, medications

Topology - Lower connectome density, efficiency, longer paths, and higher centrality was associated with a **history of psychosis** in BD, relative to those without a psychosis diagnosis.



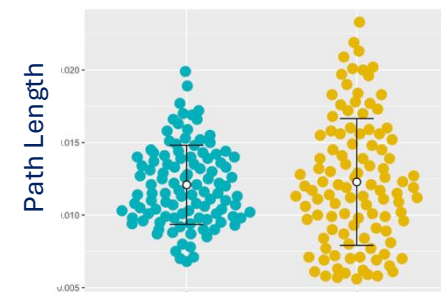
History of Psychosis

$p=.0002, d=.5$



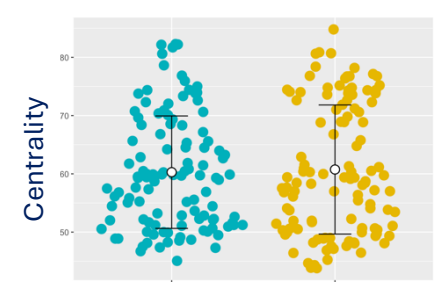
History of Psychosis

$p=.0002, d=.5$



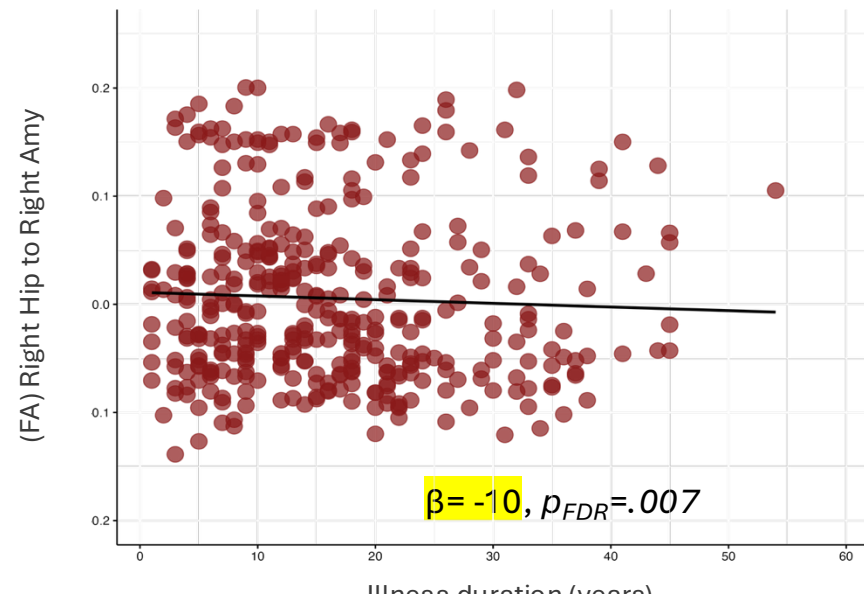
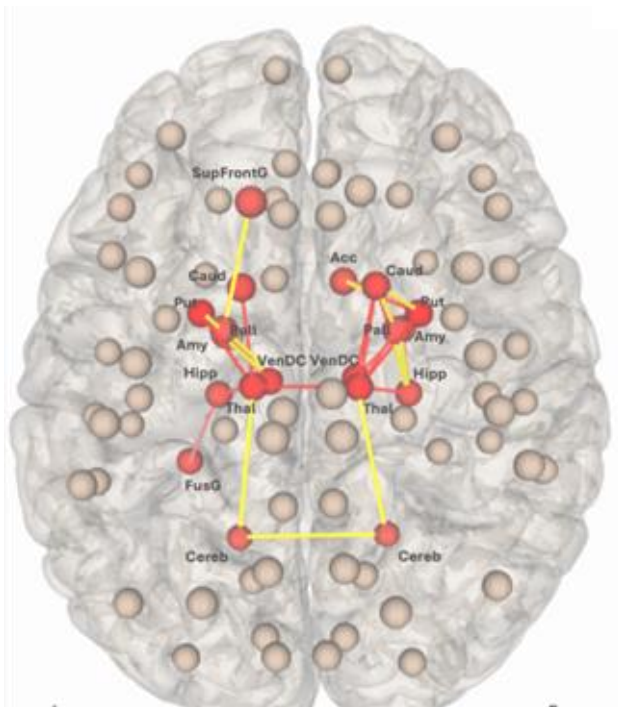
History of Psychosis

$p=.0008, d=.6$



History of Psychosis

$p=.0002, d=.6$



In BD, a **longer duration of illness** is associated with reduced connectivity between the hippocampus and amygdala (right hemisphere, FA-weighted)

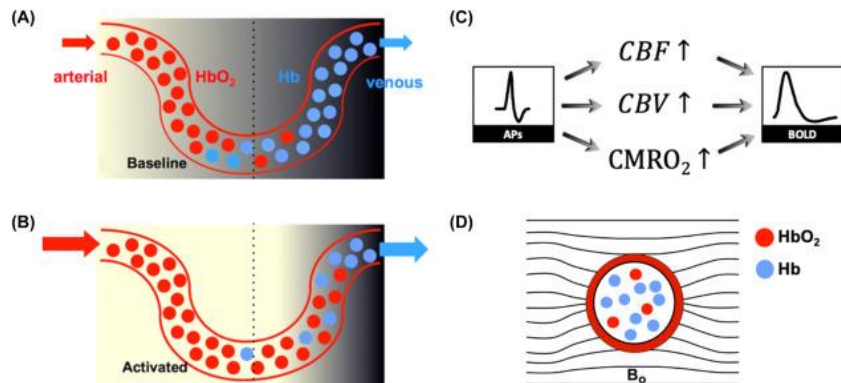
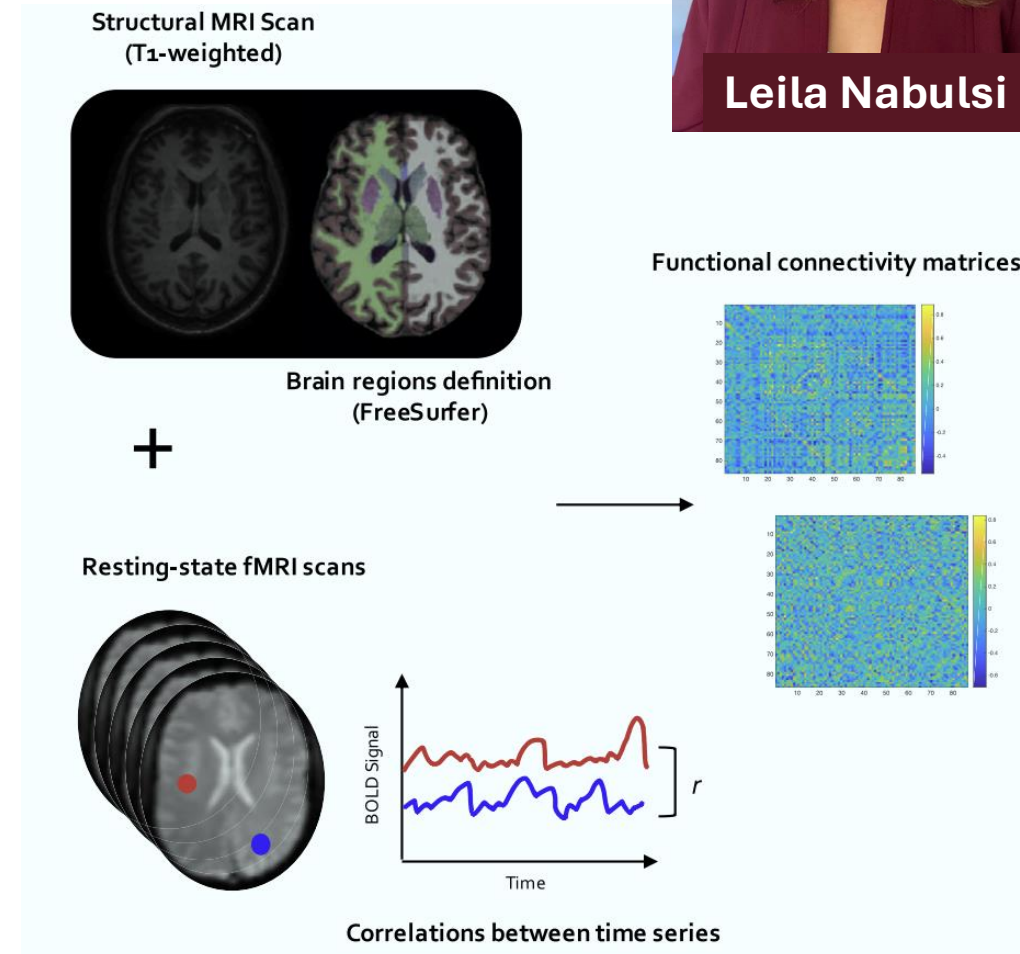
No significant association with age of illness onset, history of psychosis or number of manic or depressive episodes

Structural Connectivity - Conclusions

- **Largest study of BD connectome** to date using diffusion-weighted and structural MRI
- BD is associated with disrupted topology and reduced subnetwork connectivity in **emotion-regulatory and reward-related subsystems**
- Alterations appear to associate with some features of illness course severity
- **Large-scale multisite efforts (ENIGMA)** pooling existing neuroimaging data can enable investigation of a deeper phenotype in BD

Functional Connectivity

- Resting state function timeseries regional correlations

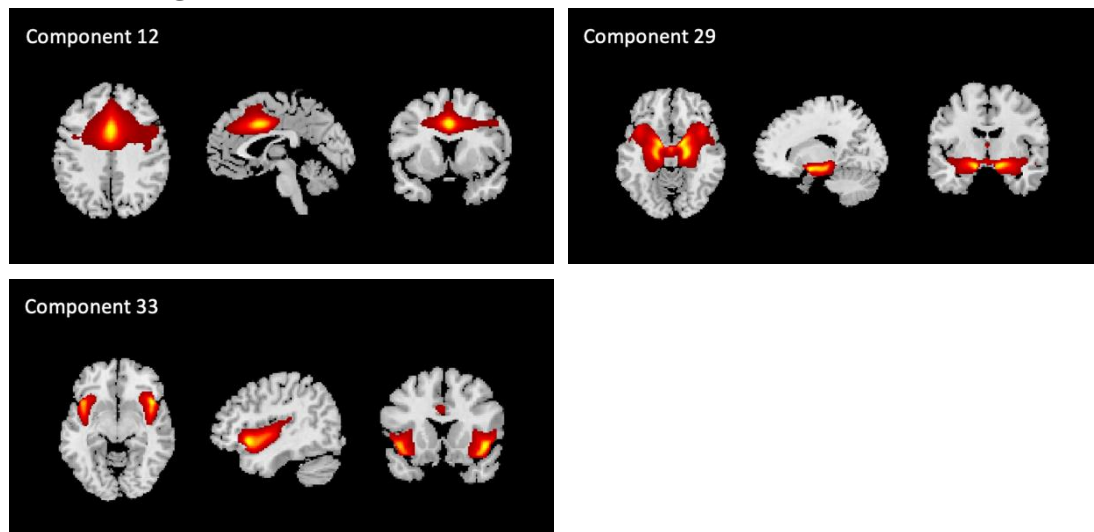


Baxter, 2023 <https://doi.org/10.1016/B978-0-323-85280-7.00002-6>

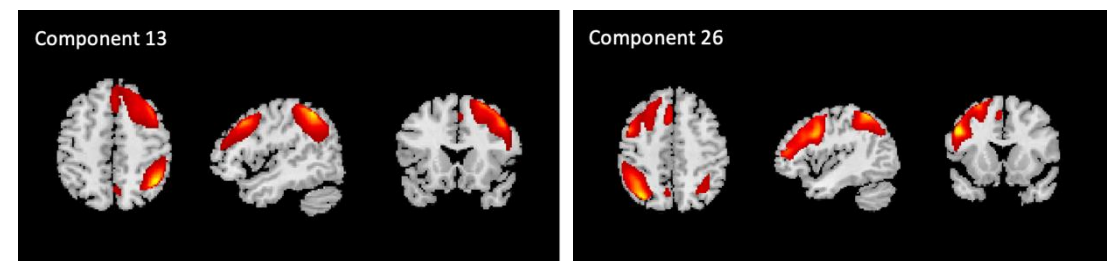
Functional Connectivity

- Resting state networks, independent component analysis (GIFT & MANCOVAN Toolbox)
- ICA: 22 noise components, 3 unreliable components ($Iq < 0.8$), 13 signal components: 6 networks
- BD relative to HC showed reduced functional connectivity in the midcingulo-insular, pericentral, lateral frontoparietal and dorsal frontoparietal networks

Midcingulo-insular Network



Lateral Frontoparietal Network

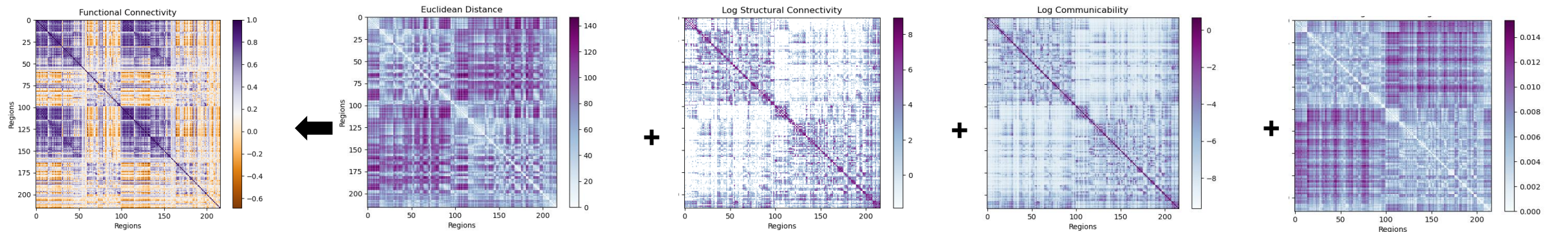


Activity in the lateral frontoparietal network associated with performance on executive function tasks (IED task) across the sample and correlated negatively within BD

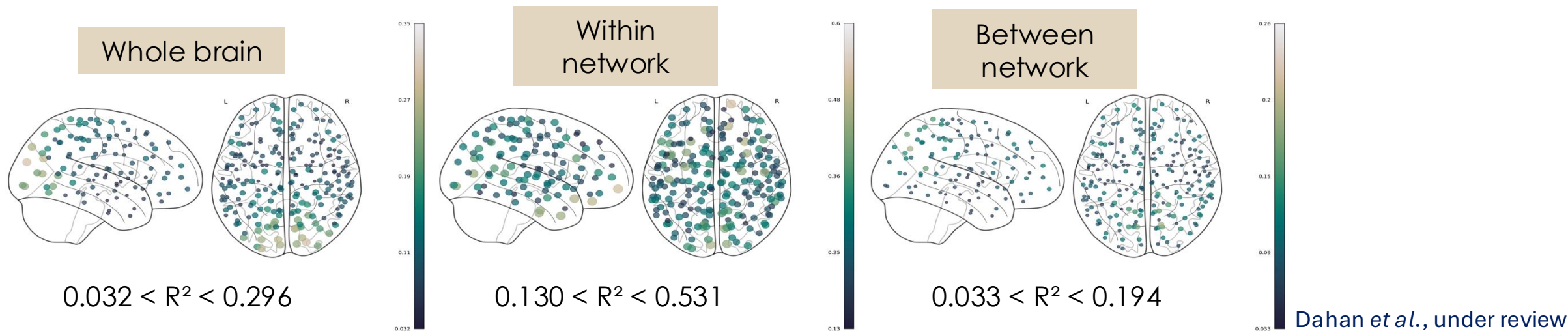
Structure - Function Coupling

Functional Connectivity

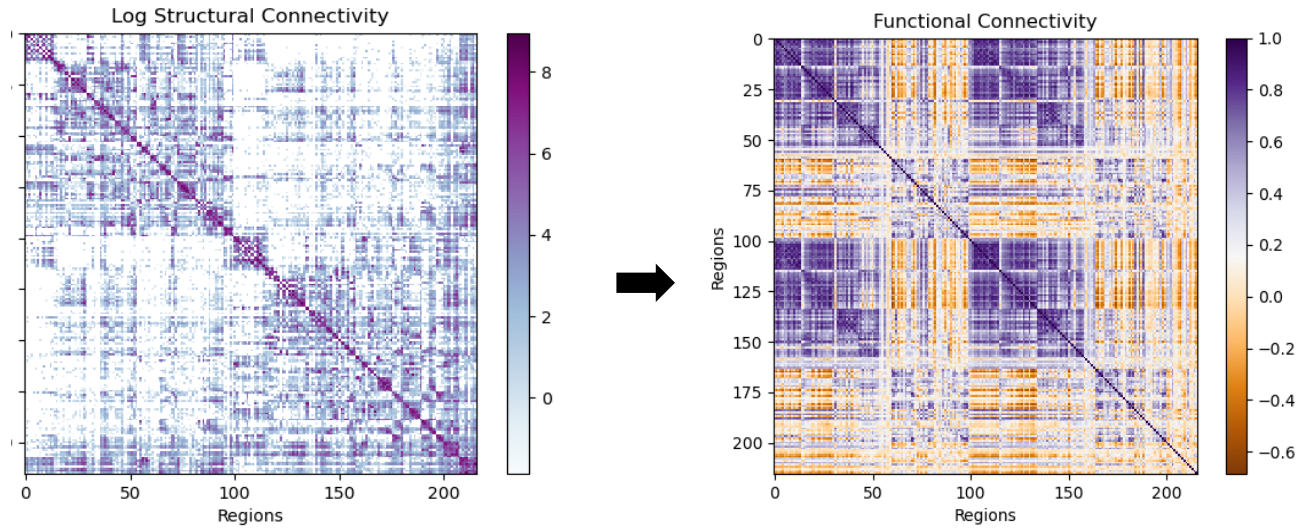
Structural Connectivity



- Estimating the degree to which structural connectivity predicts functional connectivity (adjusted R^2)
- GLM seeking main effect of diagnosis BD relative to controls, covariates motion, age, sex, site



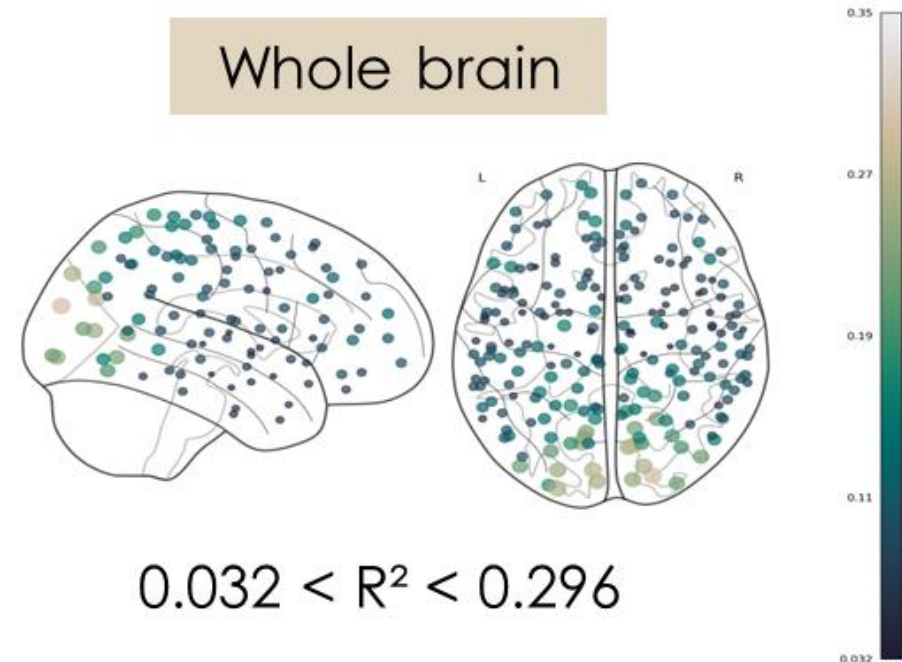
HOW DOES STRUCTURE RELATE TO FUNCTION?



- Structure-function coupling
- Cortices go from simple (unimodal) to more complex (transmodal):
 - visual cortex - unimodal, supports vision, more coupled
 - paralimbic and limbic - transmodal, less heritable, greater receptor diversification for example, less coupled

We see increased coupling in the temporal area (limbic) in bipolar disorder

Perhaps coupling is a marker of capacity to adapt functionally in a manner less constrained by anatomical connections



Structure - Function Coupling

- Consistent implication of temporal SF coupling at two parcellation resolutions in BD (n=123) relative to HC (n=326)

216 parcellation resolution:

Whole brain

Left supramarginal gyrus [SAN] ($p=0.001$, $b=-0.26$)

Left precentral gyrus [SMN] ($p=0.01$, $b=-0.24$)

Within network

Left supramarginal gyrus [SAN] ($p=0.005$, $b=-0.2$)

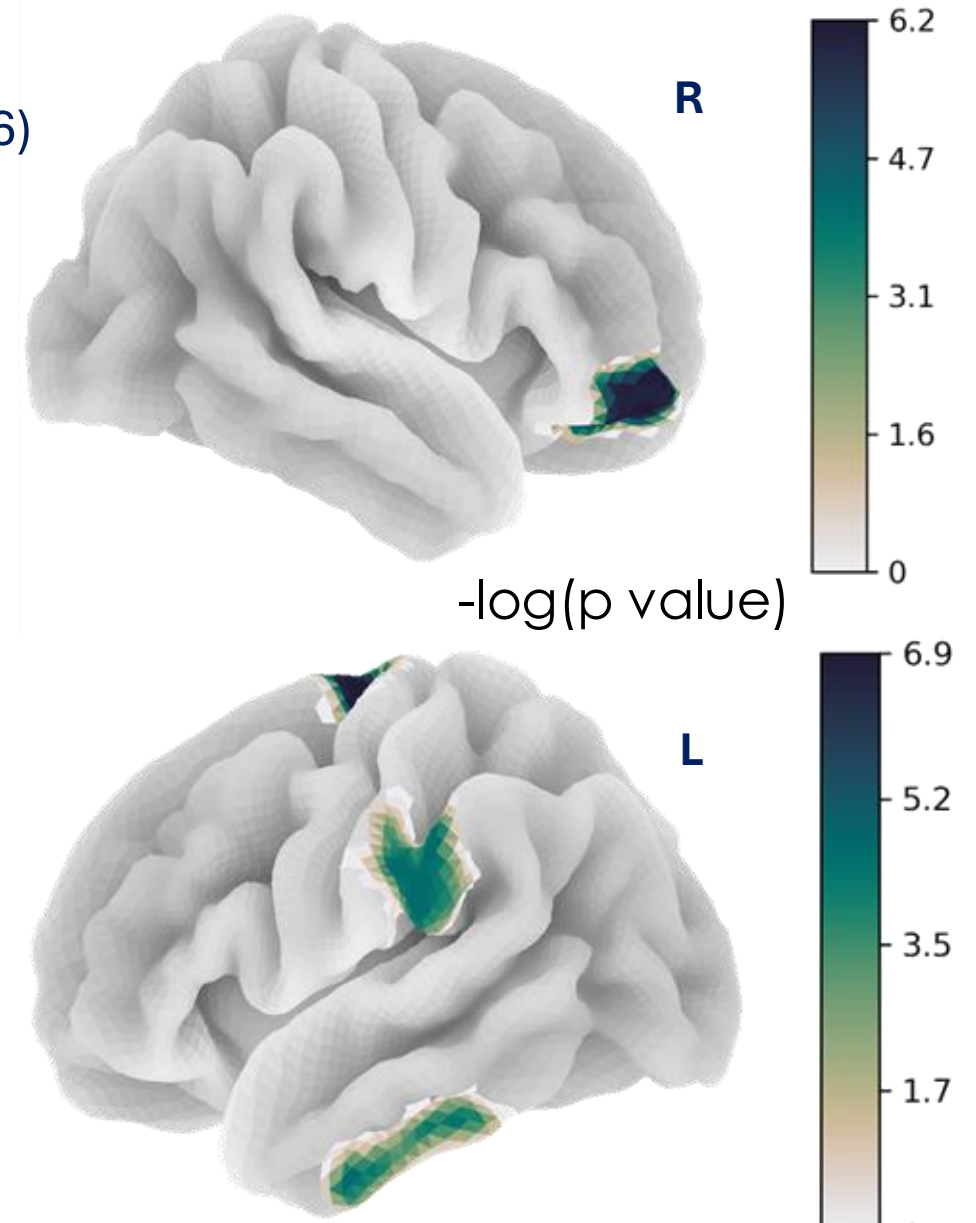
Left temporal pole [LIM] ($p=0.01$, $b=0.28$)

554 parcellation resolution:

Whole brain

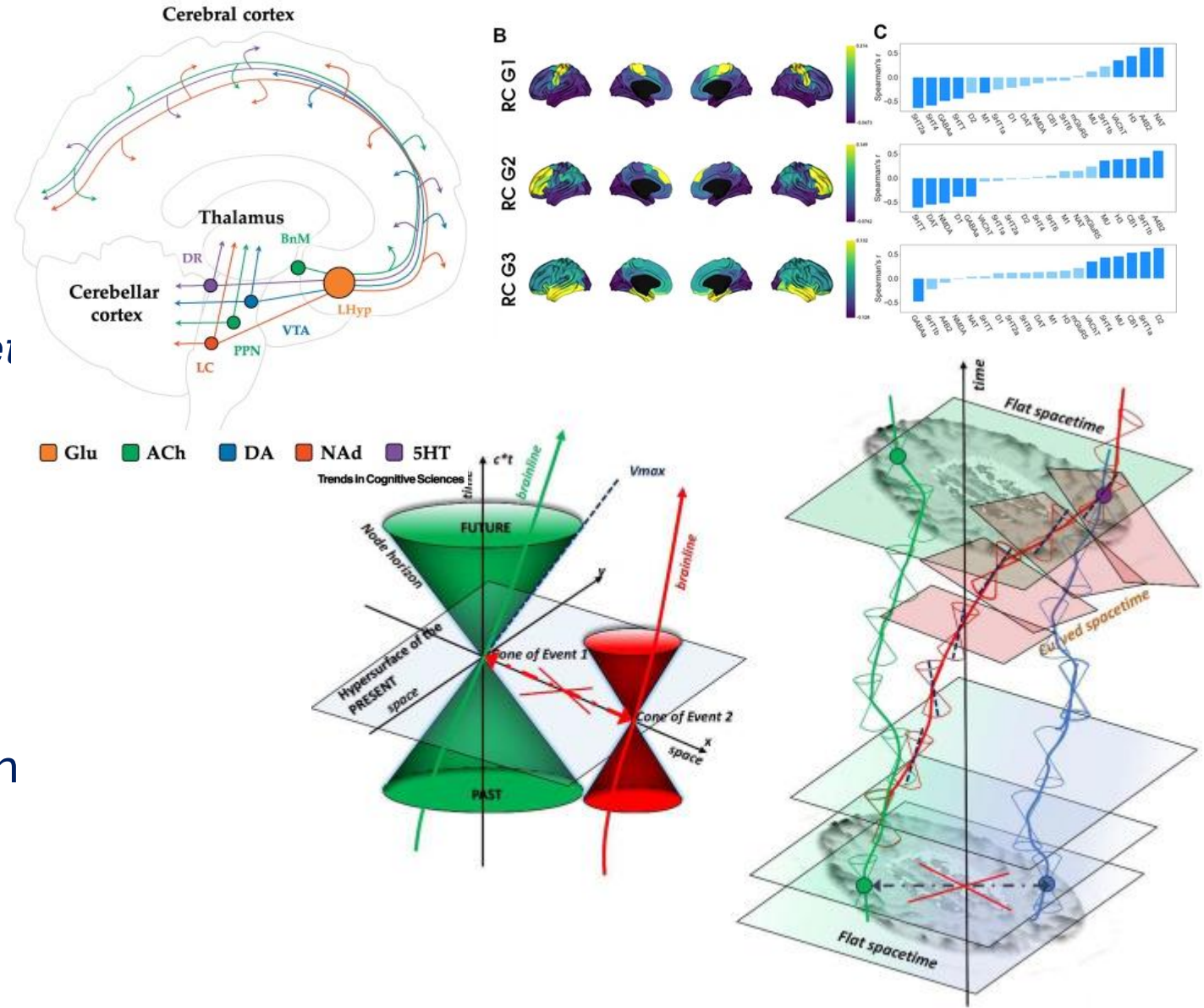
Left temporal pole [LIM] ($p=0.7$, $p^*=0.001$, $b=0.20$, same centroid coordinates)

Right frontal pole [CEN] ($p=0.002$, $b=0.28$)



Where to next for connectivity, neuroconnectomics?

- Integration with neurotransmitter system distributions and function
 - Shine, 2019 – Neuromodulation
 - DOI 10.1016/j.tics.2019.04.002
 - Hansen et al., 2022 & Hanisch et al., 2023 - Chemoarchitecture
 - DOI 10.1038/s41593-022-01186-3
 - PMID 37440423
- Some salient opportunities, potential interpretations and warnings
 - Le Bihan 2024 - Scaling the Brain
 - DOI 10.1016/j.brain.2024.100102



Thank you

To you for listening and to all the thousands of research participants, funders and past and present research team members and collaborators

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Dr Brian Hallahan
Dr Pilib O'Broin

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Dr Giulia Tronchin
Dr Theophilus Akudjedu
Dr Shane O'Connell
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